The kernel of an operating system is the core component responsible for managing hardware resources and providing essential services to other parts of the operating system and user applications. Three essential components of the OS kernel are:

- **Process Management:** The process management component is responsible for creating, scheduling, and managing processes (individual running programs) on the computer. It allocates CPU time, handles process synchronization, and manages context switching, allowing multiple processes to run concurrently on a single CPU.
- **Memory Management:** Memory management involves the allocation and control of physical and virtual memory. The kernel manages the allocation and deallocation of memory for processes, maintains memory protection, handles virtual memory addressing, and swaps data between RAM and disk when necessary to ensure efficient and secure memory usage.
- **File System Management:** File system management includes the organization, access, and maintenance of files and directories on storage devices such as hard drives or SSDs. The kernel provides a set of system calls and services that allow user applications to read, write, create, delete, and manipulate files and directories, ensuring data integrity and security.